



2Fast2Fail Final Design Report

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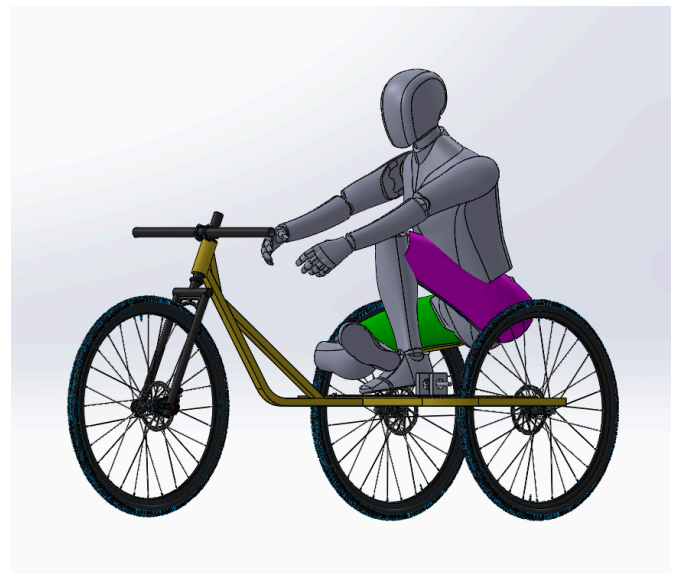


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I. Introduction and Problem Statement

The purpose of this project was to apply concepts from Materials Science, Mechanics of Solids, and Component Design to a real engineering design challenge. For this project, our team was tasked with designing and manufacturing a vehicle powered only by a Ryobi drill provided by the Component Design staff. Through this process, we applied concepts related to power transmission, material selection, structural analysis, drivetrain design, and machining.

Our team chose to compete in the endurance event, where the goal was to complete as many laps as possible on a closed course within the 30 minute allotted time. The minimum performance requirement for this event was two laps. Because endurance performance depends on both drivetrain efficiency and reliable power delivery for hills, our design prioritized torque at the driven axle over maximum speed. To achieve this, we used a larger driven sprocket on the rear axle, which increased the gear reduction, reduced the load on the drill, and allowed the drill to operate closer to a higher-efficiency RPM range.



Figure 1: (Top left to right) Back and side view of the bike; (Bottom left to right) Front and isometric view of the bike

II. Requirements

Our team designed the vehicle around several project-specific and standard competition requirements. Overall, the vehicle satisfied most of these requirements, including the drill plate size constraint, minimum manufactured component requirements, overall vehicle geometry, and the use of a three-wheel layout. However, two design requirements were not met as effectively as intended: D18, which focused on minimizing friction and maximizing efficiency, and D15, which focused on safe braking performance. These requirements are shown in Appendix II.

One of the main challenges was the drill actuation system. Our original design used a hinge mechanism connected to a brake lever on the handlebars and returned by a tension spring. During assembly and testing, this system struggled to maintain consistent cable tension and did not provide reliable drill actuation. As a result, the team replaced the hinge mechanism with a brake caliper actuation system, which provided a more consistent method of pressing the drill trigger, and was easier to fix to the drill plate.

The drivetrain also created challenges, particularly with chain tensioning. Our design used horizontal dropout tabs to tension the chain by pulling the rear wheel backward. Although this approach was expected to simplify chain tensioning, testing showed that it was difficult to adjust and maintain. Properly tensioning the chain required multiple team members, and the axle nuts did not consistently hold the wheel position over repeated use. As a result, the rear wheel gradually shifted out of alignment during operation. This issue directly affected requirement D18, which stated that “The vehicle should be designed with the least friction possible, so as to help achieve maximum distance without using any energy from the drill.” Because the rear wheel did not remain properly aligned, it developed noticeable inward camber during testing and runoff. This misalignment increased friction, reduced drivetrain efficiency, and caused the drill batteries to lose charge more quickly. At runoff, the vehicle completed 6.9 laps, which was slightly below the average for the endurance event.

Requirement D15 stated that “The bike shall have brakes to allow the driver to slow and stop the vehicle as necessary for safety and not breaking the bike.” While the vehicle included a braking system, the same cable-tensioning challenges limited the system’s effectiveness. During runoff, the brakes reduced the vehicle’s speed but did not stop it as quickly as intended. This created several safety concerns when operating near other teams and course participants.

Despite these challenges, the vehicle satisfied most of the design and competition requirements. The front-weighted layout, three-wheel configuration, drill plate constraints, and manufactured component requirements were all achieved. Future iterations should focus on improving drivetrain efficiency, rear wheel alignment, chain tensioning reliability, and braking performance.

III. Design Approach and Conceptual Design

Our design approach focused on simplicity, manufacturability, and reliability. Because the selected competition was endurance, the vehicle needed to operate consistently over multiple laps while minimizing weight, friction, and drivetrain complexity. With this in mind, our team chose a three-wheel tricycle layout with one wheel in the front and two wheels in the rear. One rear wheel would be driven by the drill-powered drivetrain, while the front wheel would provide steering and braking. The overall layout of this early concept is shown in Figure 4.

To reduce manufacturing complexity, our team used the front half of a donated bicycle, including the front fork, handlebars, front wheel, and braking system. This allowed us to

incorporate an existing steering and braking assembly rather than designing and manufacturing those systems from scratch. Since the donated bicycle also provided one rear wheel, we decided to use 26 in wheels for all three wheel locations and acquire one additional matching wheel. This helped simplify the overall geometry, cost, and kept the vehicle level.

The rider position was also selected with simplicity in mind. Instead of designing a custom seat assembly, our team used a flat wooden board as the seat platform. While this was not the most ergonomic solution, it reduced design complexity, manufacturing time, and cost. After establishing these major design decisions, the team began developing the initial tricycle frame.

The frame, shown in figure 2, used two bent tubes and three tubes with basic miter cuts to reduce the number of complex weldments and manufacturing operations. During the design process, the team also considered the required location of the drill plate, since the drill shaft and drivetrain needed to align with the driven rear wheel. Additional support tubes were later added near the head tube and beneath the rider platform to increase stiffness in higher-load areas.

To establish the approximate frame geometry, the team used CAD models from GrabCAD as reference geometry for the front bicycle assembly and general wheel placement. These models helped determine rough vehicle dimensions, frame member lengths, and rider positioning before manufacturing. Once the initial frame geometry was established, the team designed the drill plate. As shown in figure 3, the drill plate design was constrained by the maximum allowable plate size, the provided drill shaft, and the required orientation of the drill relative to the driven rear wheel. These constraints determined the approximate placement of the pillow blocks, drill mounting holes, and drivetrain alignment

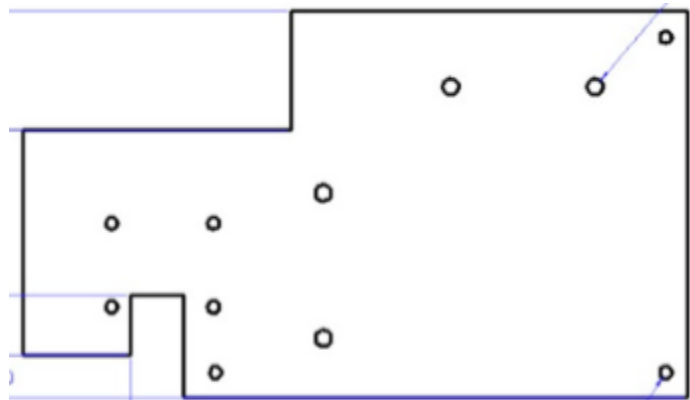


Figure 3: Initial drill plate design

The remaining components were then selected or designed around the frame and drivetrain layout. Dropouts were used to support the rear wheels and provide a method for chain tensioning. Fasteners, a wooden seat platform, and additional support hardware were selected to complete the assembly. After these components were incorporated, the team completed the initial vehicle design.



Figure 4: Overall vehicle CAD model

IV. Detailed Design Overview

The final design, shown in Figure 5, was a three-wheeled endurance vehicle with two rear wheels and one front wheel connected to the steering column. The final vehicle maintained the simple tricycle layout from the conceptual design, but several subsystems were revised after design review, manufacturing feedback, and testing. The main changes included adding gussets near the rear frame bends and head tube, redesigning the pillow blocks, reorienting the rear wheel dropouts, adding a sprocket adapter, and replacing the original drill actuation concept with a brake caliper-based system.

Several major components were modified between the initial CAD model and the final vehicle, including the frame, pillow blocks, drill shaft, drill plate, head tube, dropouts, sprocket adapter, and drill actuation system. These components were the most important to the vehicle's structural integrity, drivetrain alignment, manufacturability, and overall performance.

The first major design change involved the frame. During the design review, our team received feedback that additional gussets should be added near the rear wheel mounting weldments and the 90-degree bends in the frame. Gussets were also added near the head tube to improve stiffness and reduce the risk of weld failure. This feedback was accepted because our team had limited welding experience, and the added reinforcement improved confidence in the structural integrity of the frame. Other than these added supports, the frame design received positive feedback for being simple, manufacturable, and appropriate for the endurance competition.

The pillow block design was also revised after the design review and drawing approval process. Initially, the pillow blocks were taller and included fillets on the top edges for appearance. These features were removed to reduce manufacturing time, save material, and simplify machining. The press-fit tolerances were also corrected after feedback from the shop staff, since the original tolerance values did not match the preferred reference used for manufacturing. The final pillow block design was shorter, more compact, and better suited to support the bearings and drill shaft concentrically.

The drill shaft was a relatively simple component, with the main design change being the addition of a hex-shaped end to improve engagement with the drill chuck. The drill plate, however, required more significant revision. Our initial drill plate design exceeded the allowable material constraints, so the plate was redesigned to fit within the required dimensions while still supporting the drill, pillow blocks, shaft, and drivetrain alignment. The revised plate reduced excess overhanging material and was submitted for waterjet cutting once the final geometry was approved.

The head tube changed very little from the preliminary design. Its final dimensions were determined by measuring the donated bicycle front assembly so that the custom frame could properly interface with the existing fork and steering components. Once the correct diameter and fit were confirmed, the head tube was manufactured and incorporated into the frame.

Another important change was the orientation of the rear wheel dropouts. The initial dropout design used a vertical orientation to support the rear wheels. During design review, the team received feedback that horizontal dropouts would provide a simpler method for chain tensioning. This change eliminated the need to cut adjustment slots into the drill plate or frame and allowed the rear wheel position to be adjusted directly. Although this simplified the design, later testing showed that maintaining chain tension and rear wheel alignment was more difficult than expected.

The sprocket adapter became one of the most important late-stage design changes. At the time of the design review, our team expected to purchase a compatible sprocket commercially and did not plan to design an adapter. However, when the sprocket arrived, we realized that it was intended for a motorcycle chain and was not compatible with the bicycle chain used on our drivetrain. To solve this issue, the team rapidly designed a sprocket adapter that interfaced with a flywheel and chainring. This adapter allowed the drivetrain to use a compatible bicycle chain and was successfully manufactured before final assembly.

The drill actuation system was the most heavily iterated subsystem. The original concept used a cable-driven mechanism to pull the drill trigger. After feedback during the design review, the team changed this design to a spring-and-hinge mechanism actuated by a brake cable. During assembly and testing, this system struggled to maintain consistent cable tension and did not reliably press the drill trigger. After several attempts to improve the mechanism, the team replaced it with a discarded brake caliper. The brake caliper provided a more consistent actuation method and was mounted using additional threaded rods. Once tested, this became the final drill actuation system used on the vehicle. With these changes incorporated, the final CAD model of the vehicle is shown in Figure 5.



Figure 5: Final CAD iteration of the bike

The final iteration of the bike included all of the above changes, as well as a few last-minute fixes that are highlighted in the images below. Most notably, with little time left, the overhanging portion of the drill plate was secured with a copious amount of zip ties. The orientation of the drill plate was awkward once the chain had been fully tensioned, and this appeared to be the only option that did not require further frame alteration. The fully manufactured version of the bike is shown below.

On the day of the run-off, our vehicle was one of the first to be tested in the endurance competition. After struggles with battery transfer and incorrect drill settings, our vehicle ultimately made it 6.9 laps around the designated course. We competed against one other team in the same timeslot, and their bike appeared to go much faster than ours. Regardless, our design was able to meet and exceed the minimum requirements for a satisfactory grade, and the team was happy with the overall performance. A notable issue we faced was



Figure 6: Painted and fully assembled bike frame

the changing of the drill battery once each one had been fully exhausted, as we did not bring our charger outside and the location of it was quite far from the designated pit stops. Additionally, the second battery was not fully charged when it was needed, and we imagine we could have made it to at least 7 full laps had this not been the case.

After the competition, our vehicle metrics were determined. The final length of the bike was determined to be just under 7 feet,, with the weight reporting around 46 lbs. These values meant that we had met the required design constraints, and we did not lose any points from exceeding them.

V. Component Analysis

A. Critical Components

Part I: Drill Plate

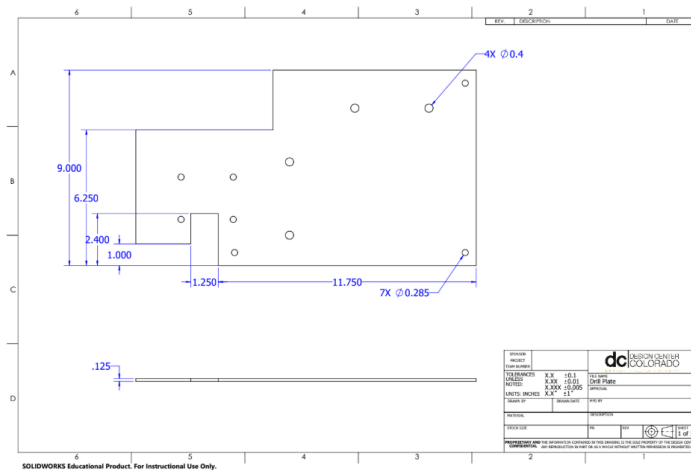


Figure 7: (Left) SolidWorks Drawing of the final Drill Plate design
(Right) Manufactured drill plate

The drill plate was one of the most critical components of our design simply because it was our means of securing the drill to our bike frame, allowing it to maintain steady, consistent actuation of the left wheel. Our understanding of gear ratios, material selection, shafts, bearings, and fasteners influenced several aspects of our design.

The plate was designed with precise hole placement to ensure proper alignment between the drill chuck and the drive shaft, minimizing any moment loads that could arise from misalignment. We selected aluminum due to its machinability and sufficient yield strength to withstand vibration forces. Aluminum was also the best choice since it would be inexpensive to create several iterations with the material, since we were undoubtedly going to make changes at some point during the design and testing process. Fastener placement and spacing were also carefully considered to prevent any single fastener from experiencing a disproportionate share of the load. The plate we designed balances compactness and rigidity, ensuring the drill remains secure under dynamic loading conditions.

Part II: Head Tube

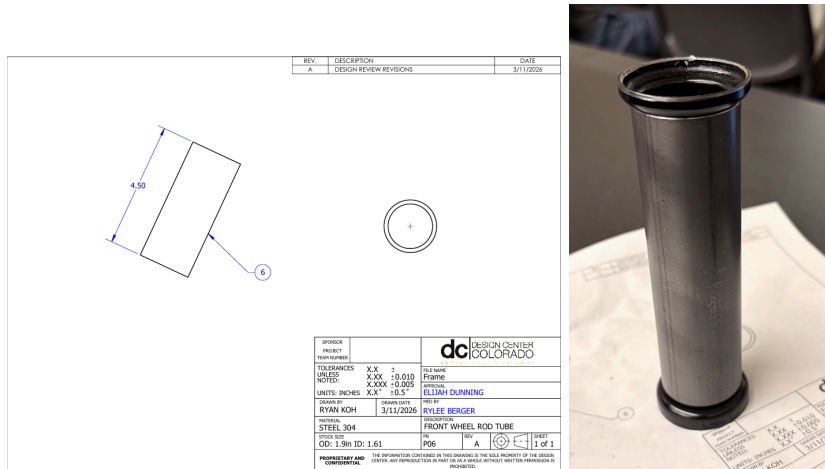


Figure 8: *LEFT Head Tube Drawing; *RIGHT Manufactured Head Tube

The head tube serves as one of the most structurally critical components of the frame, as it is our means of connecting the rest of the frame to the assembly of the bike's front wheel. This transfers both the bending moments and axial loads generated while the bike is in motion. We used the tube caps from the bike we used for parts to determine the size and geometry of the tube, while selecting a material that could handle our combined loading conditions. We selected steel 304 for the head tube based on its well-characterized mechanical properties, including a yield strength of approximately 215 MPa and ease of welding, an essential factor given that the head tube had to be joined to the rest of the frame. Steel 304 also offered sufficient ductility to absorb impact loads without brittle fracture, a necessity for the conditions of our use case. From a stress analysis perspective, the wall thickness was sized to keep the maximum bending stress well below the material's yield strength under worst conditions, providing an adequate factor of safety while keeping the component, light, cheap and machinable.

B. Failure Analysis

Based on the failure calculations we did on our bike frame, we determined that the head tube is the most likely to fail. Other factors of the frame, such as the seat beams, had a safety factor of about 22.068. The head tube had the lowest safety factor, around 3.248, so it was determined that failure would most likely occur at this point.

C. Final Gear Ratio

Originally, our team planned to use a 4.5:1 gear ratio. However, during the preliminary design review, we received feedback that this ratio may be too low for the torque demands of our vehicle. Based on this feedback, our team increased the final gear ratio to 5:1. This was achieved using a 50-tooth driven sprocket and a 10-tooth driver sprocket. The larger gear ratio increased the torque delivered to the drive wheel while still maintaining a reasonable output speed for the endurance competition. The torque being for the last $\frac{1}{4}$ of the competition hill and the speed for the straight sections. For our drill characterization, a graph of the speed vs torque can be found in the Appendix.

VI. Manufacturing & Assembly

All of our custom manufactured parts were made with materials we got from the Idea Forge. Most of the parts that we manufactured were not driving parts of our design. We mainly designed them to fit wherever they needed to go on the bike. We were not able to manufacture some of the parts ourselves, so we designed the timeline around this fact to speed up the testing of our design.

The drill plate was manufactured using a water jet. We did not participate in the manufacturing of this component, because the Idea Forge staff are the only ones allowed to use the water jet. Due to the fact that we had no control over the time it would take to make this part, we got our drawing submitted early for manufacturing. We wanted the drill plate so we could work on tensioning the chain and the actuation, so this was a driving part of the design.

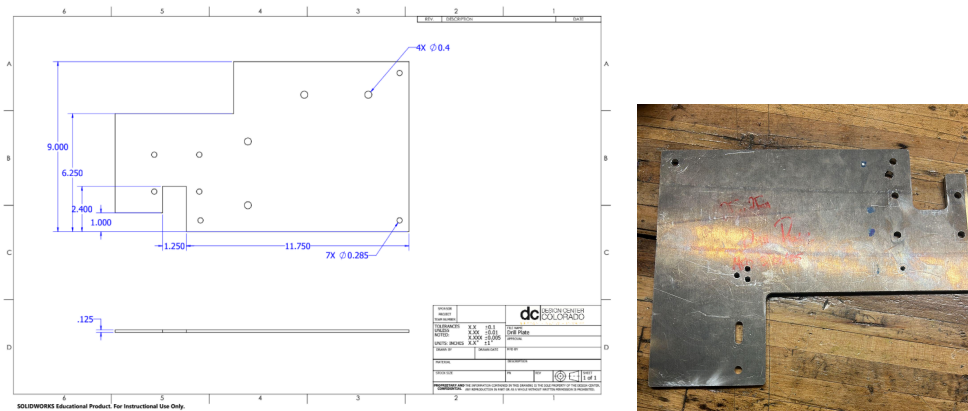


Figure 10: *LEFT Drill Plate CAD Drawing; *RIGHT Manufactured Drill Plate

The pillow blocks were made from material given to us by the Idea Forge in a mill. We made the mounting holes on the pillow block by manually controlling the machine. However, the bearing holes that the drill shaft went through were made with a CNC program on the mill. This is because we did not have a drill bit of the same size as the bearing holes. We made a mistake while machining one of our pillow blocks, because we made a 0.003” mistake on the bearing hole. While making the initial bearing hole, we accidentally lowered the mill instead of raising it up, which made the initial cut deeper than it should have been. We didn’t end up having to redo that part and it did not change any part of our design.

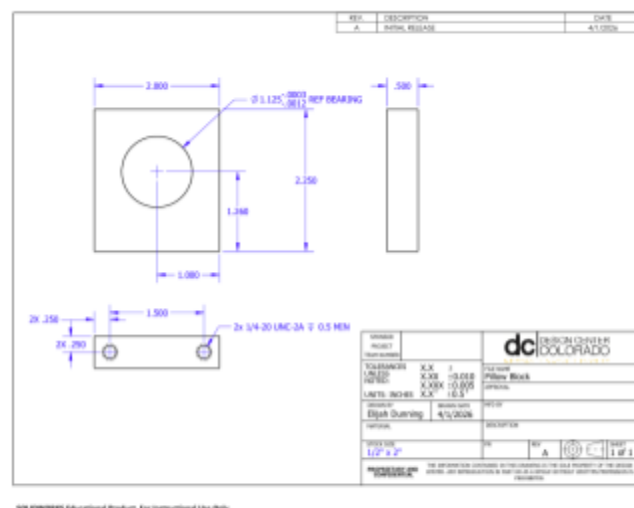


Figure 10. Pillow Block SolidWorks Drawing

The drill shaft was made with a lathe, also with material from the Idea Forge. We laid the drill shaft material with its axial axis along the x-axis and then used the lathe to make a hexagonal shape 1" into the shaft material. This component was easy to make and there were no mishaps. It was one of the least time-consuming parts.

The sprocket adapter was also made in a water jet. We found a flywheel on Amazon. Unfortunately, we were pressed for time with this component, because we did not think it was necessary for our competition. Once it was made aware to us that we needed to have it, we designed and sent this part for manufacturing the same day, which was also the last day that students were allowed to submit waterjet requests. This did not affect our design in any way because we were already behind schedule for manufacturing and assembly.



Figure 11. Final Manufactured Blocks with Bearings

To make the head tube, we started by choosing a tube using the diameter of the caps. We then used the mill to cut away most of the material, and the lathe to refine and soften the edges.

For our sprocket, we started by using the lathe to make sure its teeth fit snugly inside the narrowest part of the chain. We then used a mill to add two concentric holes with threading for the

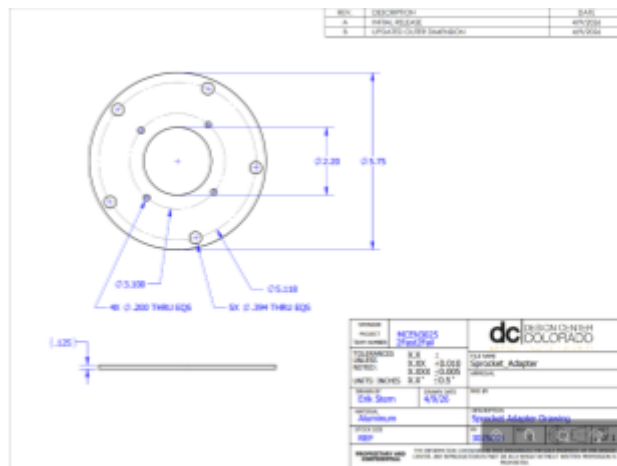


Figure 12. Sprocket Adapter SolidWorks Drawing

set screws. We initially forgot to add divots (“dimples”) at the point of contact between the set screws and the drill shaft. This meant that the drill shaft initially would not spin the drill, because it would just slide against the drill shaft. We removed the set screws and used a drill to create indents in



Figure 15: Side View of Machined Sprocket

the drill shaft. This allowed the drill to drive the sprocket.

The most difficult part to manufacture was, surprisingly enough, the pillow blocks. It took about three hours to do because the Idea Forge staff were quite busy, and we needed to use the CNC function on the mill. It was more time-consuming than difficult.

Manufacturing and assembly took much longer than expected for a multitude of reasons. We initially planned to have all manufacturing and assembly done between March 30th and April 3rd, but we ended up finishing the assembly of our bike on April 17th. We then tested it on the 17th and the 20th. This took longer than we expected mainly due to poor scheduling of team members. Many team members were not available to manufacture during the originally agreed upon times, leading to other team members having to find extra time in their schedules to work on the manufacturing.



Figure 16. Front view of manufactured sprocket

VII. Bill of Materials & Total Cost

Key components of our assembled vehicle were sourced from four primary suppliers, along with several donated or university-provided materials. During the initial planning stage, minimizing overall cost was a major design objective. To achieve this, we prioritized using existing materials, donated bicycle components, and stock materials available through the university whenever possible.

Several materials for key components were provided at no cost by CU Boulder, which significantly reduced our total budget. These included the drill plate material, pillow block material, drill shaft material, sprocket adapter/dropouts, head tube, and multiple fasteners ($\frac{1}{4}$ -20 and $\frac{3}{8}$ -20 bolts, nuts, and related hardware). Having free access to these materials allowed us to avoid purchasing many small but necessary structural and assembly components.

Additional savings came from a donated bicycle frame and parts provided by Ryan. These donated components included the braking system, handlebars, front frame section, and two of the three wheels used in the final vehicle. Reusing these parts reduced both cost and fabrication time.

Commercially purchased components were primarily sourced from McMaster-Carr, which supplied several drivetrain and mechanical hardware items. These purchases included a 10-tooth steel plain bore sprocket, high-strength flange nuts, extension springs, and two $\frac{1}{2}$ -inch shaft ball bearings, totaling **\$51.91**. Structural tubing for the chassis was purchased from Metal Supermarkets, where three 80-inch sections of mild steel square tube were acquired for **\$54.96**. Additional purchased components included a 45-tooth rear sprocket from Faster Minis for **\$17.65**, a plywood panel from Home Depot for **\$9.18**, and a 26-inch bicycle wheel purchased through Facebook Marketplace for **\$29.95**.

The final total project cost was **\$161.62**, which remained comfortably below the \$200 project budget limit. This result was somewhat unexpected, as our initial estimate placed the

project much closer to the maximum allowable budget. The largest reason for this reduction was receiving fasteners and raw materials at no cost after the original estimate had been completed. In the original Components Off-The-Shelf (COTS) Bill of Materials, several of these parts were assumed to be purchased. Once these no-cost materials became available, the final budget decreased substantially.

VIII. Timeline

The timeline above differed from the real schedule our team followed, mainly because the original

plan was made before we fully understood the time required for purchasing, drawing approval, manufacturing, and assembly. Ideally, all materials would have been acquired before spring break, but delivery schedules, exams, and

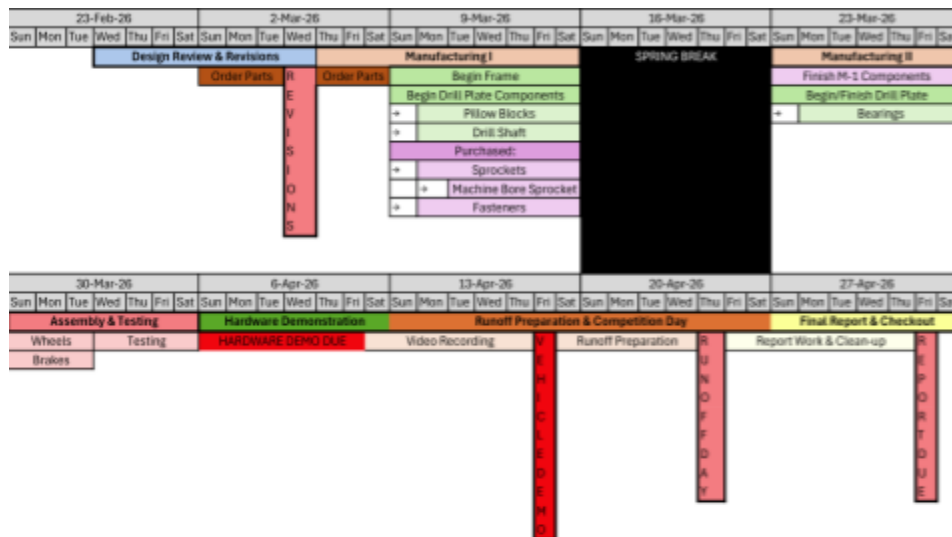


Figure 17. The initial timeline developed for our design review

repeated shop redline revisions delayed the start of manufacturing until March 25th. As a result, the week originally planned for assembly and testing was instead used to finish manufacturing the remaining components.

Despite this delay, our team completed all required components before the hardware demonstration. Assembly began during the week of April 13th, when we encountered issues with drill actuation and chain tensioning. These problems delayed testing and forced us to record the vehicle demonstration video on the due date. However, the vehicle still met the minimum distance and weight requirements. The remaining time before runoff was used to improve reliability and prepare the vehicle for competition. Overall, the main deviation from the original timeline was the delayed manufacturing schedule caused by part procurement, drawing approval, and assembly challenges.

IX. Conclusion

The most significant lessons our team learned throughout this project involved adaptability, communication, planning, and the realities of manufacturing. While our group began the semester organized through a shared Google Drive folder and clear task lists, we

initially struggled with knowing who was responsible for each task and when deliverables were due. This highlighted the importance of active communication rather than relying solely on shared documents. As the semester progressed, our weekly lab meetings and more consistent communication through our group chat greatly improved coordination and accountability.

One of the largest challenges occurred during the design phase of this project. Our team felt confident in the initial concept and expected only minor revisions following the design review. However, we discovered that the drill plate exceeded dimensional regulations, despite having previously received approval. This required a complete redesign within 24 hours so the part could still be manufactured on schedule. That experience demonstrated how quickly constraints can change in engineering and how important it is to remain flexible under pressure.

During the manufacturing cycle, we also learned the value of schedule buffers. Our team intentionally created an aggressive timeline that included two extra contingency days in case unexpected issues arose. Those additional days ultimately became necessary. For example, one chassis tube was bent to the wrong angle, requiring us to shorten and modify it while still preserving acceptable driver ergonomics. This reinforced the reality that fabrication rarely proceeds exactly as planned, and successful teams must be prepared to troubleshoot quickly. The final two weeks of the project presented another major challenge when our original actuation system did not perform as intended. Rather than forcing an unreliable solution, our team rapidly redesigned the mechanism and implemented a simpler, more dependable alternative before runoff testing. This was one of the most rewarding moments of the project because it showed our ability to respond to failure with practical engineering problem solving.

From a teamwork perspective, the biggest lesson was learning how to work effectively with different personalities and work styles. Some team members preferred structured planning, while others worked best through hands-on iteration. Early in the project this created friction, especially when responsibilities overlapped between CAD, fabrication, and assembly. Over time, we improved by communicating expectations more clearly and respecting how each person contributed best.

If this project were extended into a second semester, several design changes would be prioritized. First, we would improve driver ergonomics by redesigning the seating position for greater comfort during endurance events. Second, we would redesign the rear wheel alignment and camber geometry, as excessive camber reduced efficiency, speed, and available power transfer. Third, we would create a more robust fixturing system for mounting the drill motor and drivetrain components. Finally, we would continue refining the actuation mechanism to further improve consistency and reliability.

From a process standpoint, we would also implement more formal internal design reviews and manufacturing readiness checks. These reviews would focus on validating calculations, confirming dimensions, and ensuring the logic of each subsystem before fabrication begins. Additional coordination would also be placed on tasks with overlapping ownership, particularly between CAD design and manufacturing.

Overall, this project required our team to design, manufacture, assemble, and validate a functional human-powered/drill-powered vehicle while working within cost, time, and rules constraints. Along the way, we encountered setbacks, design changes, and manufacturing issues that tested our teamwork and problem-solving ability. Despite these challenges, we successfully delivered a high quality completed and competitive vehicle under budget and gained valuable hands-on experience in engineering design, fabrication, communication, and iteration. More than

anything, this project demonstrated that successful engineering is not about avoiding problems but rather about solving them efficiently as a team.

APPENDIX

II. Design Requirements Figures

ID	Category	Specification statement	Source	Date	Verification Type	Progress	Comments
D15	Design	The bike shall have brakes to allow the driver to slow and stop the vehicle as necessary for safety and not breaking the bike. Looking at bike brakes.	Ourselves		Do the brakes work		Must decide between caliper vs. disk brakes (~+\$10)

Figure 1: Design Requirement 15

ID	Category	Specification statement	Source	Date	Verification Type	Progress	Comments
D18	Design	The vehicle should be designed with the least friction possible, so as to help achieve maximum distance without using any energy from the drill	Ourselves	28-Jan-26	Test or Measurement		

Figure 2: Design Requirement 18

VII. Bill of Materials and Total Cost

Table 1. Bill of Materials

Part Number	Item Description	Quantity	Stock Material	Material Cost	Material Vendor	Responsible Drafter	Responsible Engineer	Received?
3025-A00-PM01	Drill Plate Material	1	Aluminum	\$0.00	CU Boulder	N/A	Leighanne Bennett	Yes
3025-A00-PM02	Pillow Block Material	1	Aluminum	\$0.00	CU Boulder	N/A	Leighanne Bennett	Yes
3025-A00-PM03	Drill Shaft Material	1	1566 ground shaft	\$0.00	CU Boulder	N/A	Leighanne Bennett	Yes
3025-A00-PM04	Sprocket Adapter and/or dropouts	As needed	A36 Steel	\$0.00	CU Boulder	N/A	Rylee Berger	Yes
3025-A	Head Tube	1	A36	\$0.00	CU	N/A	Rylee	Yes

00-PM 05			Steel		Boulder		Berger	
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Table 2. Bill of Commercial-Off-The-Shelf Parts

	Description	Quantity	Unit Cost	Total Cost	Vendor	Responsible Engineer	Received ?
3025-A 00-PC0 1	Brakes	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A 00-PC0 2	Handle bars	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A 00-PC0 3	Front Frame	1	\$0.00	\$0.00	Ryan Koh	Ryan Koh	Yes
3025-A 00-PC0 4	80" Mild Steel Square Tube Structural (Welded)	3	\$18.32	\$54.96	Metal Supermarket	Karl Kubala	Yes
3025-A 00-PC0 5	Bike Wheels 26"	3	\$29.95	\$29.95	REI	Elijah Dunning	Yes
3025-A 00-PC0 6	1/4-20 3/4" Nuts/Bolt	4	\$0.00	\$0.00	Andy - CU Boulder	Erik Stern	Yes
3025-A 00-PC0 7	10 Tooth Steel Plain Bore Sprocket (6793K141)	1	\$18.16	\$18.16	McMaster-Carr	Erik Stern	Yes
3025-A 00-PC0 8	High Strength Wide Flange Nuts (1/4) (91627A115)	1	\$0.66	\$0.66	McMaster-Carr	Erik Stern	Yes
3025-A 00-PC0 9	5pk Steel Extension Spring (5108N56)	1	\$3.53	\$17.65	McMaster-Carr	Erik Stern	Yes
3025-A 00-PC1 0	45 Tooth Rear Sprocket	1	\$15.62	\$15.62	Faster Minis	Erik Stern	Yes
3025-A 00-PC1 1	Steel 1/2" Shaft Ball Bearings (60355K173)	2	\$7.72	\$15.44	McMaster-Carr	Erik Stern	Yes

3025-A 00-PC1 2	1/4-20 1" Nuts/Bolt	4	\$0.00	\$0.00	Andy - CU Boulder	Rylee Berger	Yes
3025-A 00-PC1 3	1/4-20 1.625" Nuts/Bolts	3	\$0.00	\$0.00	Andy - CU Boulder	Elijah Dunning	Yes
3025-A 00-PC1 4	1/4 in. x 2 ft. x 2 ft. Sanded Plywood Project Panel	1	\$9.18	\$9.18	Home Depot	Rylee Berger	Yes

V. Component Analysis

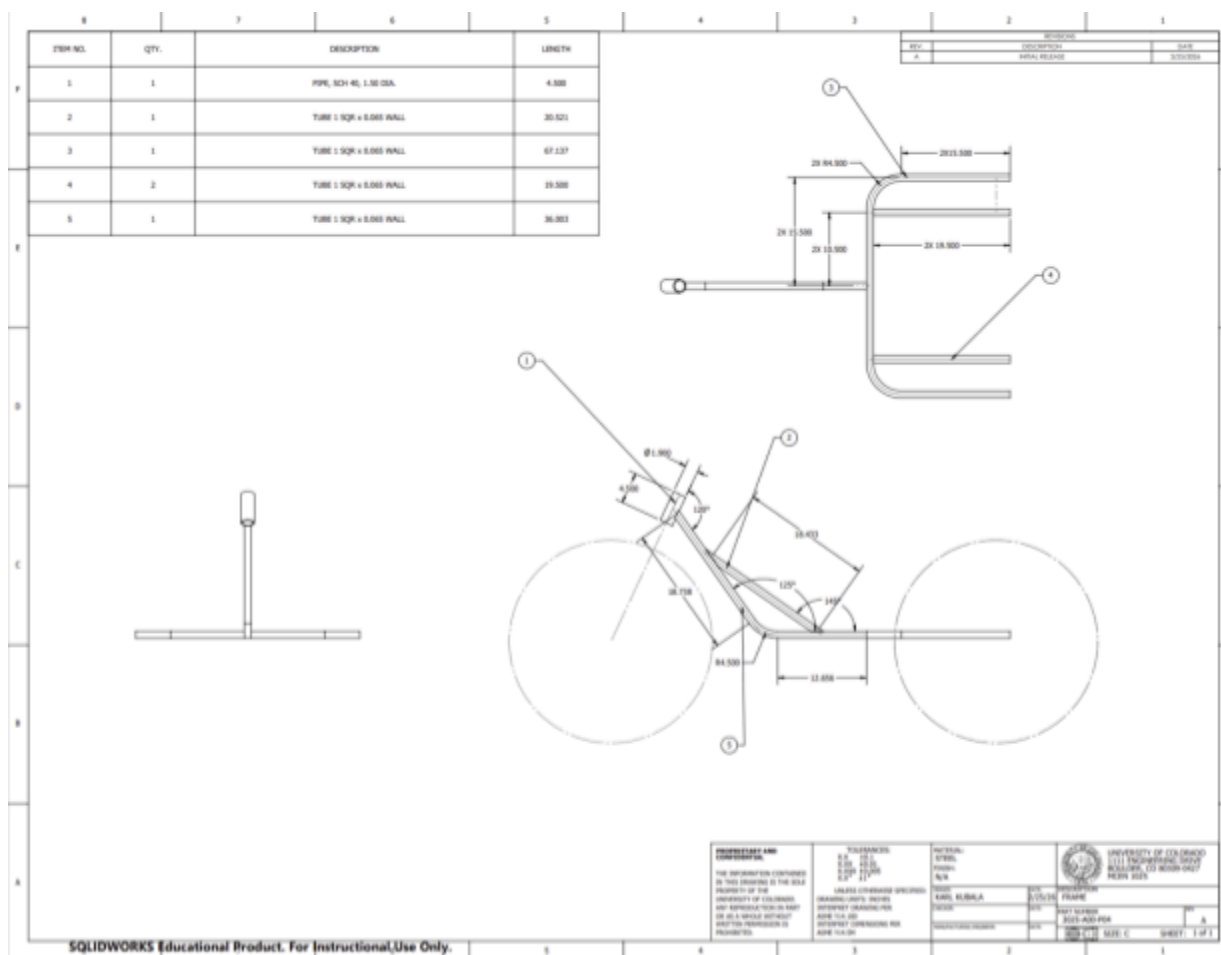


Figure 9: Full Frame CAD drawing

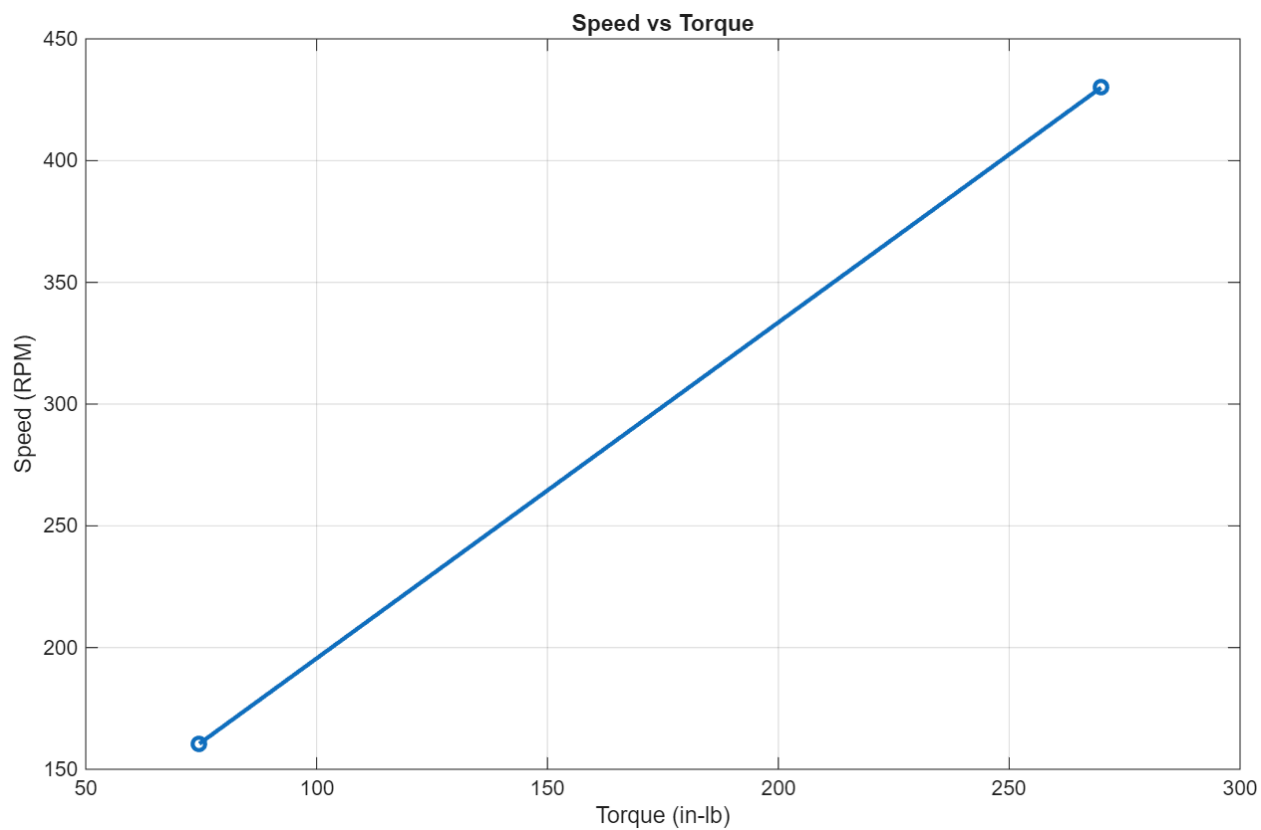


Figure 10: Graph of speed (RPM) vs torque (in-lb) for our unique drill characterization

FAILURE ANALYSIS CALCULATIONS BEGIN ON NEXT
PAGE